

LAPORAN PRAKTIKUM  
WORKSHOP KECERDASAN BUATAN



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TRM A 24 (SEMESTER 5)

PROGRAM STUDI SARJANA TERAPAN TEKNOLOGI REKAYASA MULTIMEDIA

JURUSAN TEKNOLOGI MULTIMEDIA KREATIF

POLITEKNIK ELEKTRONIKA NEGERI SURABAYA

2026

➤ OUTPUT BAGIAN EXAMPLE

## 1. SVM.PY

```
[Running] python -u "C:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\example1-SVM.PY"
```

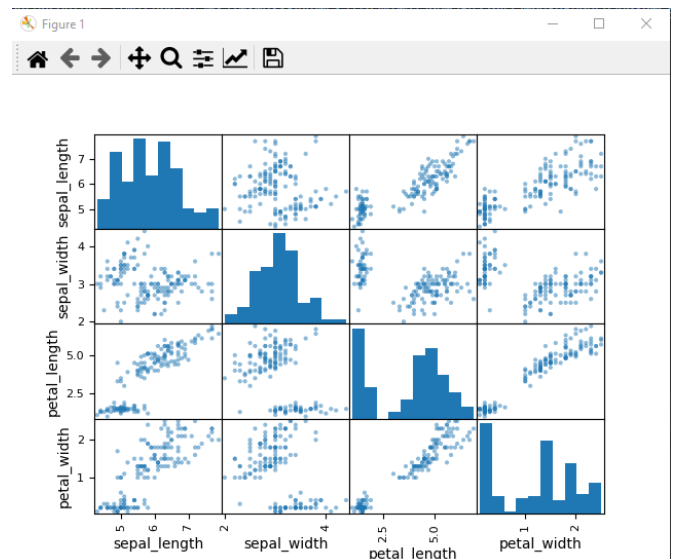
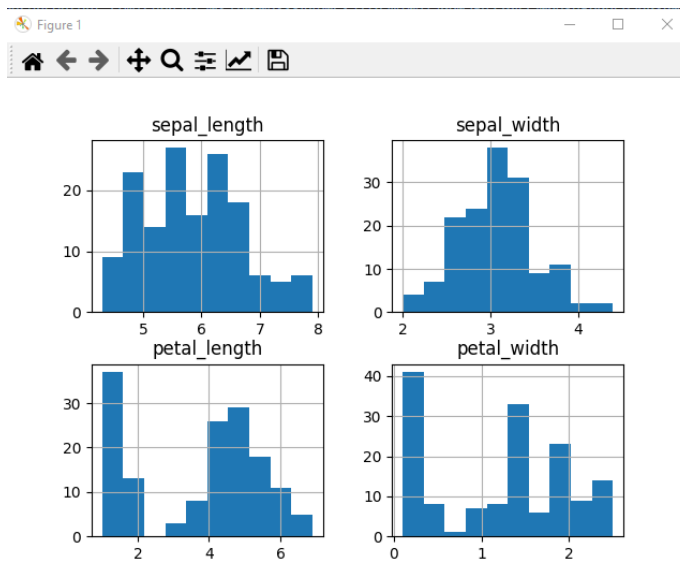
## 2. SVM2.PY

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\example2-SVM2.PY"  
[0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0  
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0  
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1  
2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2  
2 2]  
[0]  
  
[Done] exited with code=0 in 2.368 seconds
```

### 3. SVM3.PY

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\example3-SVM.PY"  
[0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0  
0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1  
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2  
2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2  
2 2]  
[0]  
  
[Done] exited with code=0 in 2.469 seconds
```

#### 4. SVM4



```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS 55\AI\BAB 3\EXAMPLE\example4-SVM4.PY"
(150, 5)
|   sepal_length  sepal_width  petal_length  petal_width  species
0           5.1           3.5           1.4           0.2  setosa
1           4.9           3.0           1.4           0.2  setosa
2           4.7           3.2           1.3           0.2  setosa
3           4.6           3.1           1.5           0.2  setosa
4           5.0           3.6           1.4           0.2  setosa
5           5.4           3.9           1.7           0.4  setosa
6           4.6           3.4           1.4           0.3  setosa
7           5.0           3.4           1.5           0.2  setosa
8           4.4           2.9           1.4           0.2  setosa
9           4.9           3.1           1.5           0.1  setosa
|   sepal_length  sepal_width  petal_length  petal_width  species
140          6.7           3.1           5.6           2.4  virginica
141          6.9           3.1           5.1           2.3  virginica
142          5.8           2.7           5.1           1.9  virginica
143          6.8           3.2           5.9           2.3  virginica
144          6.7           3.3           5.7           2.5  virginica
145          6.7           3.0           5.2           2.3  virginica
146          6.3           2.5           5.0           1.9  virginica
147          6.5           3.0           5.2           2.0  virginica
148          6.2           3.4           5.4           2.3  virginica
149          5.9           3.0           5.1           1.8  virginica
|   sepal_length  sepal_width  petal_length  petal_width
count    150.000000    150.000000    150.000000    150.000000
mean      5.843333     3.054000     3.758667     1.198667
std       0.828066     0.433594     1.764420     0.763161
min       4.300000     2.000000     1.000000     0.100000
25%      5.100000     2.800000     1.600000     0.300000
50%      5.800000     3.000000     4.350000     1.300000
75%      6.400000     3.300000     5.100000     1.800000
max       7.900000     4.400000     6.900000     2.500000
0
species
setosa      50
versicolor  50
virginica   50
dtype: int64
[0]

[Done] exited with code=0 in 166.015 seconds
```

## 5. CANCER.PY

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS 55\AI\BAB 3\EXAMPLE\example5-CANCER.PY"
['mean radius' 'mean texture' 'mean perimeter' 'mean area'
 'mean smoothness' 'mean compactness' 'mean concavity'
 'mean concave points' 'mean symmetry' 'mean fractal dimension'
 'radius error' 'texture error' 'perimeter error' 'area error'
 'smoothness error' 'compactness error' 'concavity error'
 'concave points error' 'symmetry error' 'fractal dimension error'
 'worst radius' 'worst texture' 'worst perimeter' 'worst area'
 'worst smoothness' 'worst compactness' 'worst concavity'
 'worst concave points' 'worst symmetry' 'worst fractal dimension']
[[1.799e+01 1.038e+01 1.228e+02 ... 2.654e-01 4.601e-01 1.189e-01]
 [2.057e+01 1.777e+01 1.329e+02 ... 1.860e-01 2.750e-01 8.902e-02]
 [1.969e+01 2.125e+01 1.300e+02 ... 2.430e-01 3.613e-01 8.758e-02]
 ...
 [1.660e+01 2.808e+01 1.083e+02 ... 1.418e-01 2.218e-01 7.820e-02]
 [2.060e+01 2.933e+01 1.401e+02 ... 2.650e-01 4.087e-01 1.240e-01]
 [7.760e+00 2.454e+01 4.792e+01 ... 0.000e+00 2.871e-01 7.039e-02]]
['malignant' 'benign']

[Done] exited with code=0 in 2.176 seconds
```

## 6. CANCER2.PY

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\example6-CANCER2.PY"
is_cancer (0)      predicted_cancer  predicted_healthy
is_healthy (1)      0                66

Classification Report:

              precision    recall  f1-score   support

     0       1.00        0.83        0.91         48
     1       0.89        1.00        0.94         66

 accuracy          0.93         114
 macro avg         0.95         114
weighted avg         0.94         114

[Done] exited with code=0 in 3.501 seconds
```

## 7. NAIVEBAYES.PY

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\example7- NAIVEBAYES.PY"
[[5.1 3.5 1.4 0.2]
 [4.9 3.  1.4 0.2]
 [4.7 3.2 1.3 0.2]
 [4.6 3.1 1.5 0.2]
 [5.  3.6 1.4 0.2]
 [5.4 3.9 1.7 0.4]
 [4.6 3.4 1.4 0.3]
 [5.  3.4 1.5 0.2]
 [4.4 2.9 1.4 0.2]
 [4.9 3.1 1.5 0.1]
 [5.4 3.7 1.5 0.2]
 [4.8 3.4 1.6 0.2]
 [4.8 3.  1.4 0.1]
 [4.3 3.  1.1 0.1]
 [5.8 4.  1.2 0.2]
 [5.7 4.4 1.5 0.4]]
```

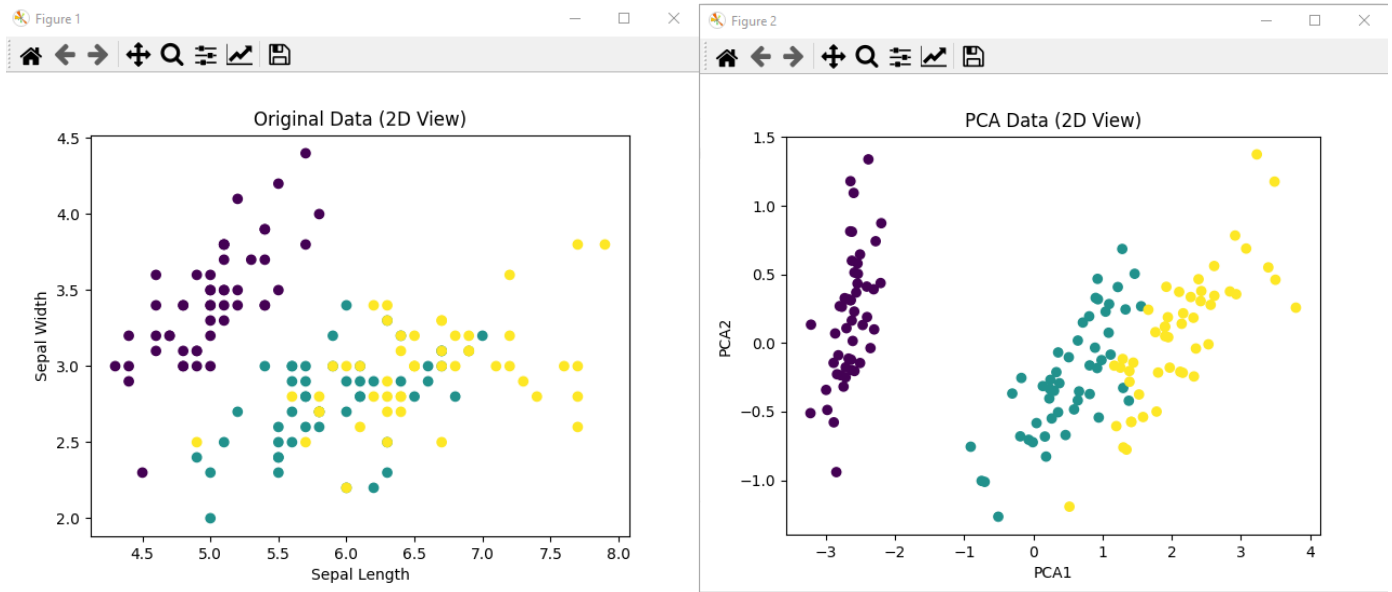
## 8. LDA.PY

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\example8- LDA.PY"
[[-1.66853167 -1.29901346  0.2746472  -0.60362044]
 [-2.9728827  -1.08878294  0.70885958  0.42281857]
 [-0.59614125 -1.37007001 -3.11685659  0.64445203]
 ...
 [ 0.91711204  1.10596645  0.86766522 -2.25625012]
 [ 0.10027664  1.45875846 -0.44360274 -0.67002328]
 [ 1.0415229  -0.01987143  0.15216419 -1.9405334  ]]
[1]
```

## 9. THE PCA.PY

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\example9- THE PCA.PY"
[[-1.66853167 -1.29901346  0.2746472  -0.60362044]
 [-2.9728827  -1.08878294  0.70885958  0.42281857]
 [-0.59614125 -1.37007001 -3.11685659  0.64445203]
 ...
 [ 0.91711204  1.10596645  0.86766522 -2.25625012]
 [ 0.10027664  1.45875846 -0.44360274 -0.67002328]
 [ 1.0415229  -0.01987143  0.15216419 -1.9405334  ]]
[0.35054565 0.26955436 0.1920725  0.1878275  ]
[42.31488145 37.10598956 31.32228453 30.97422379]
```

## 10. THE PCA 2



## 11. DT.PY

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\example11- DT.PY"
Total points: 75 Correctly labeled points : 72

[Done] exited with code=0 in 3.338 seconds
```

## 12. RF.PY

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\example12- RF.PY"
Total points: 75 Correctly labeled points : 70

[Done] exited with code=0 in 2.947 seconds
```

## 13. KNN.PY

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\example13- KNN.PY"
Total points: 75 Correctly labeled points : 72

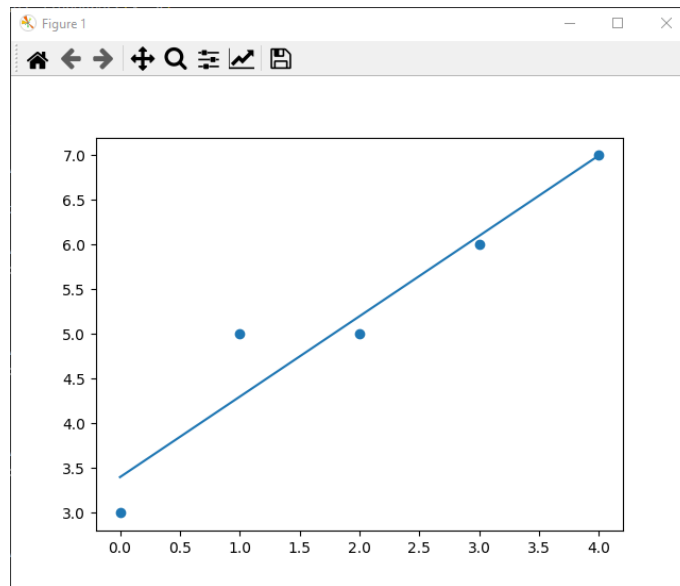
[Done] exited with code=0 in 2.442 seconds
```

## 14. CLASSIFY.PY

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\example14- CLASSIFY.PY"
SVM: 0.9466666666666667
Naive Bayes: 0.9466666666666667
LDA: 0.96
QDA: 0.96
Decision Tree: 0.96
Random Forest: 0.96
Nearest Neighbors: 0.96
Neural Networks: 0.9733333333333334

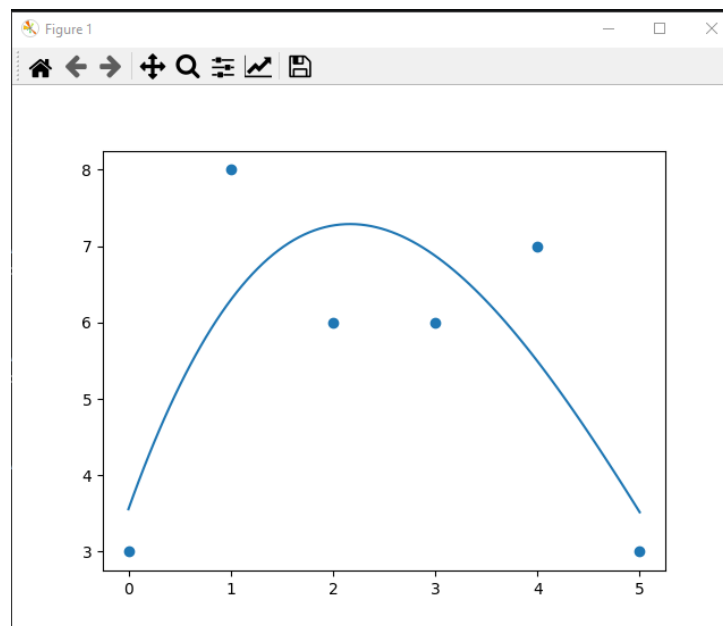
[Done] exited with code=0 in 3.221 seconds
```

## 15. LINEARR.PY



```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\example15- LINEARR.PY"
slope: 0.9
intercept: 3.4000000000000004
[Done] exited with code=0 in 39.538 seconds
```

## 16. POLYRR.PY



```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\example16- POLYR.PY"
| | | 3 2
0.06481 x - 1.075 x + 3.749 x + 3.556
[Done] exited with code=0 in 50.137 seconds
```

## 17. LSF.PY

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\example17- LSF.PY"
c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\example17- LSF.PY:9: RuntimeWarning: overflow encountered in exp
| return a * np.exp(- b * x) + c
Best fit a b c: [ 4.09564212e+02  6.21466356e-02 -3.04106531e+02]
Best fit covariance: [[ 4.92882349e+05 -8.73845292e+01 -4.95819953e+05]
| [-8.73845292e+01  1.55361498e-02  8.79409202e+01]
| [-4.95819953e+05  8.79409202e+01  4.98815648e+05]]

[Done] exited with code=0 in 1.338 seconds
```

## 18. MLR.PY

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\example18- MLR.PY"
Coefficients:
| [0.3 0.3 0.3]
Intercept:
| 0.9999999999999999
Prediction:
| [7.9]

[Done] exited with code=0 in 2.619 seconds
```

## 19. LOGR.PY

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\example19- LOGR.PY"
[31]
[[0.00111396 0.00810048 0.03285548 0.09767681 0.61033535 0.24991792]]
0.6666666666666666

[Done] exited with code=0 in 2.183 seconds
```

## 20. KMEANS.PY

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\example20- KMEANS.PY"
[1 1 1 0 0 0]
[[10.      2.      3.      ]
| [ 1.      2.      2.66666667]]
[0]

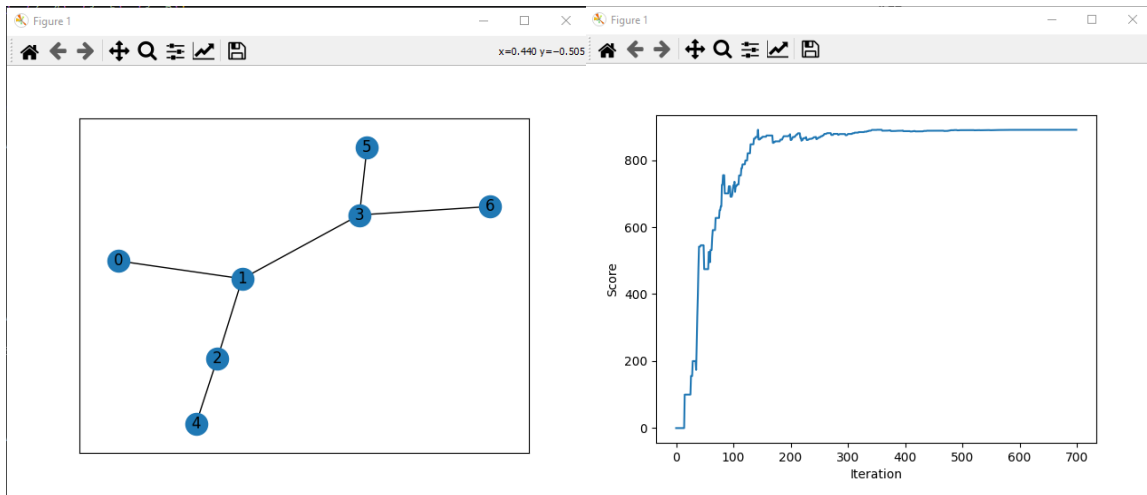
[Done] exited with code=0 in 2.822 seconds
```

## 21. SEMIL.PY

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\example21- SEMIL.PY"
[ 0. -1. -1. -1. -1. -1. -1.  1.]
[0. 0. 0. 0. 1. 1. 1. 1.]

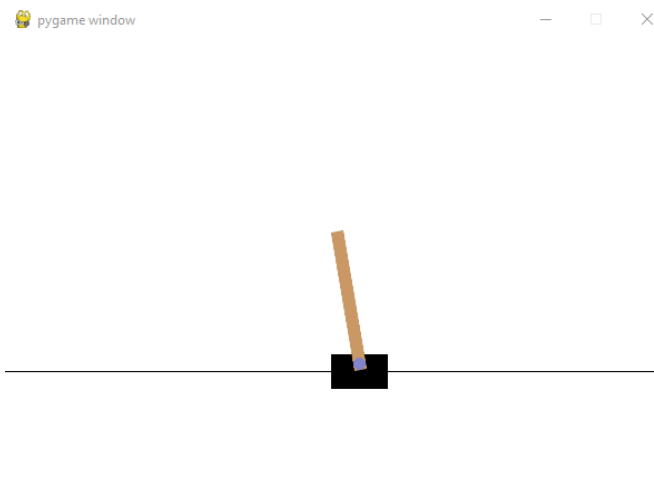
[Done] exited with code=0 in 3.471 seconds
```

## 22. TES.PY



```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS 55\AI\BAB 3\EXAMPLE\example22- TEST.PY"
(0, 1)
(1, 2)
(1, 3)
(2, 4)
(3, 5)
(3, 6)
[[ -1.  0. -1. -1. -1. -1. -1.]
 [ 0. -1.  0.  0. -1. -1. -1.]
 [-1.  0. -1. -1.  0. -1. -1.]
 [-1.  0. -1. -1. -1.  0. 100.]
 [-1. -1.  0. -1. -1. -1. -1.]
 [-1. -1. -1.  0. -1. -1. -1.]
 [-1. -1. -1.  0. -1. -1. 100.]]
max_value 0.0
Score: 0
max_value 0.0
Score: 0
max_value 0.0
Score: 0
max_value 0.0
```

## 23. GYM.PY

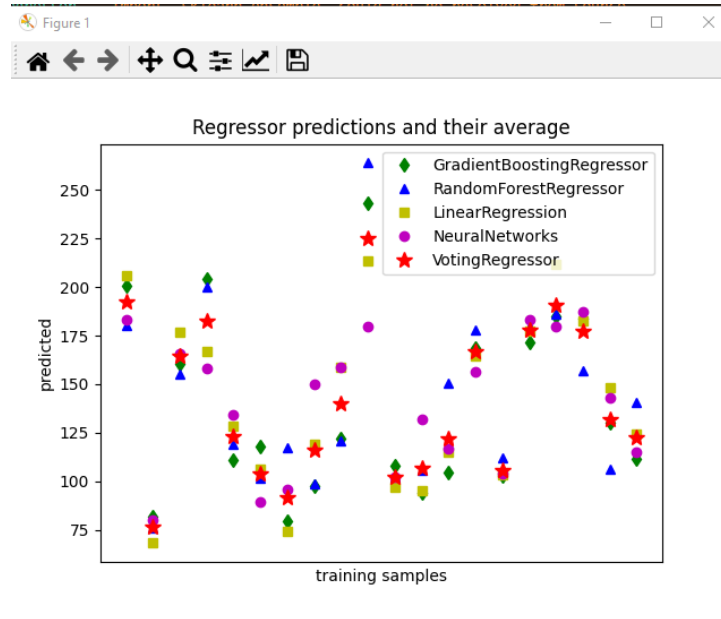


## 24. ENSEMBLE.PY

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\example24- ENSEMBLE1.PY"
Accuracy: 0.95 (+/- 0.04) [Logistic Regression]
Accuracy: 0.94 (+/- 0.04) [Random Forest]
Accuracy: 0.91 (+/- 0.04) [naive Bayes]
Accuracy: 0.95 (+/- 0.04) [SVM]
Accuracy: 0.95 (+/- 0.04) [Ensemble]

[Done] exited with code=0 in 3.749 seconds
```

## 25. ENSEMBLE2.PY



```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\example25- ENSEMBLE2.PY"
C:\Users\rifqy\AppData\Local\Programs\Python\Python311\Lib\site-packages\sklearn\neural_network\_multilayer_perceptron.py:691: ConvergenceWarning:
Stochastic Optimizer: Maximum iterations (1000) reached and the optimization hasn't converged yet.
  warnings.warn(
C:\Users\rifqy\AppData\Local\Programs\Python\Python311\Lib\site-packages\sklearn\neural_network\_multilayer_perceptron.py:691: ConvergenceWarning:
Stochastic Optimizer: Maximum iterations (1000) reached and the optimization hasn't converged yet.
  warnings.warn(
VotingRegressor(estimators=[('gb', GradientBoostingRegressor(random_state=1)),
                             ('rf', RandomForestRegressor(random_state=1)),
                             ('lr', LinearRegression()),
                             ('NN',
                              MLPRegressor(max_iter=1000, random_state=1))])

[Done] exited with code=0 in 52.913 seconds
```

## 26. AUTOML.TEST.PY

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\example26- AUTO ML TEST.PY"

Adjusted R-Squared ... Time Taken
Model ...
LarsCV 0.404730 ... 0.024766
LassoCV 0.403655 ... 0.117985
LassoLarsCV 0.403633 ... 0.060938
OrthogonalMatchingPursuitCV 0.401335 ... 0.018366
ExtraTreesRegressor 0.400309 ... 0.312472
Lasso 0.398528 ... 0.011530
LassoLars 0.398516 ... 0.024937
PoissonRegressor 0.392079 ... 0.028723
```

## 27. PYCARET\_DEMO.PY

|       |                                 |        |        |        | Model  | Accuracy | AUC | Recall | Prec. | \ |
|-------|---------------------------------|--------|--------|--------|--------|----------|-----|--------|-------|---|
| lr    | Logistic Regression             | 0.9718 | 0.0000 | 0.9718 | 0.9780 |          |     |        |       |   |
| knn   | K Neighbors Classifier          | 0.9718 | 0.9830 | 0.9718 | 0.9780 |          |     |        |       |   |
| qda   | Quadratic Discriminant Analysis | 0.9718 | 0.0000 | 0.9718 | 0.9780 |          |     |        |       |   |
| lda   | Linear Discriminant Analysis    | 0.9718 | 0.0000 | 0.9718 | 0.9780 |          |     |        |       |   |
| nb    | Naive Bayes                     | 0.9445 | 0.9868 | 0.9445 | 0.9525 |          |     |        |       |   |
| et    | Extra Trees Classifier          | 0.9445 | 0.9935 | 0.9445 | 0.9586 |          |     |        |       |   |
| gbc   | Gradient Boosting Classifier    | 0.9355 | 0.0000 | 0.9355 | 0.9416 |          |     |        |       |   |
| dt    | Decision Tree Classifier        | 0.9264 | 0.9429 | 0.9264 | 0.9502 |          |     |        |       |   |
| rf    | Random Forest Classifier        | 0.9264 | 0.9909 | 0.9264 | 0.9343 |          |     |        |       |   |
| ada   | Ada Boost Classifier            | 0.9155 | 0.0000 | 0.9155 | 0.9401 |          |     |        |       |   |
| ridge | Ridge Classifier                | 0.8227 | 0.0000 | 0.8227 | 0.8437 |          |     |        |       |   |
| svm   | SVM - Linear Kernel             | 0.7618 | 0.0000 | 0.7618 | 0.6655 |          |     |        |       |   |
| dummy | Dummy Classifier                | 0.2864 | 0.5000 | 0.2864 | 0.0822 |          |     |        |       |   |

|     | F1     | Kappa  | MCC    | TT (Sec) |
|-----|--------|--------|--------|----------|
| lr  | 0.9712 | 0.9573 | 0.9609 | 0.699    |
| knn | 0.9712 | 0.9573 | 0.9609 | 0.431    |
| qda | 0.9712 | 0.9573 | 0.9609 | 0.009    |
| lda | 0.9712 | 0.9573 | 0.9609 | 0.009    |
| nb  | 0.9438 | 0.9161 | 0.9207 | 0.010    |
| et  | 0.9426 | 0.9161 | 0.9246 | 0.047    |

## 28. LAZY PREDICT

- A

```

... --- 5 Data Pertama ---
      sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)
0                5.1             3.5             1.4             0.2
1                4.9             3.0             1.4             0.2
2                4.7             3.2             1.3             0.2
3                4.6             3.1             1.5             0.2
4                5.0             3.6             1.4             0.2

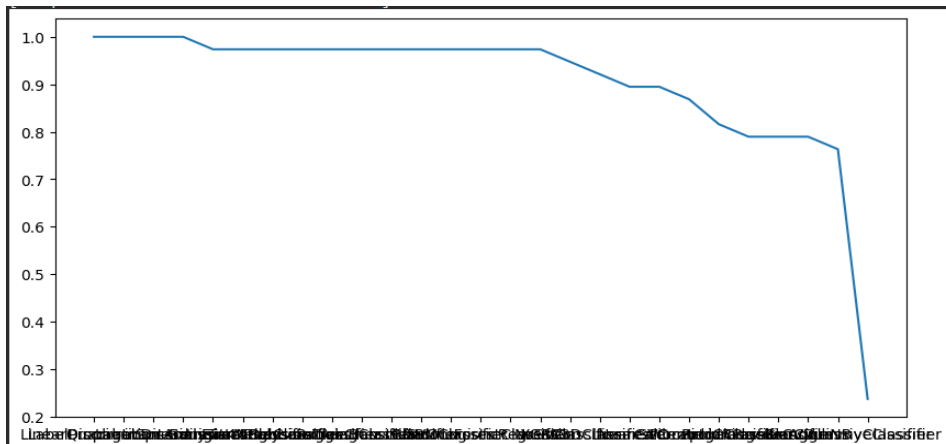
--- Membandingkan Model (Seperti PyCaret) ---
      Model  Accuracy  Std Dev
Logistic Regression  0.973333  0.024944
K-Nearest Neighbors  0.973333  0.024944
SVM                0.966667  0.021082
Gradient Boosting   0.960000  0.032660
Decision Tree       0.960000  0.032660
Random Forest       0.960000  0.024944
Extra Trees         0.953333  0.033993

```

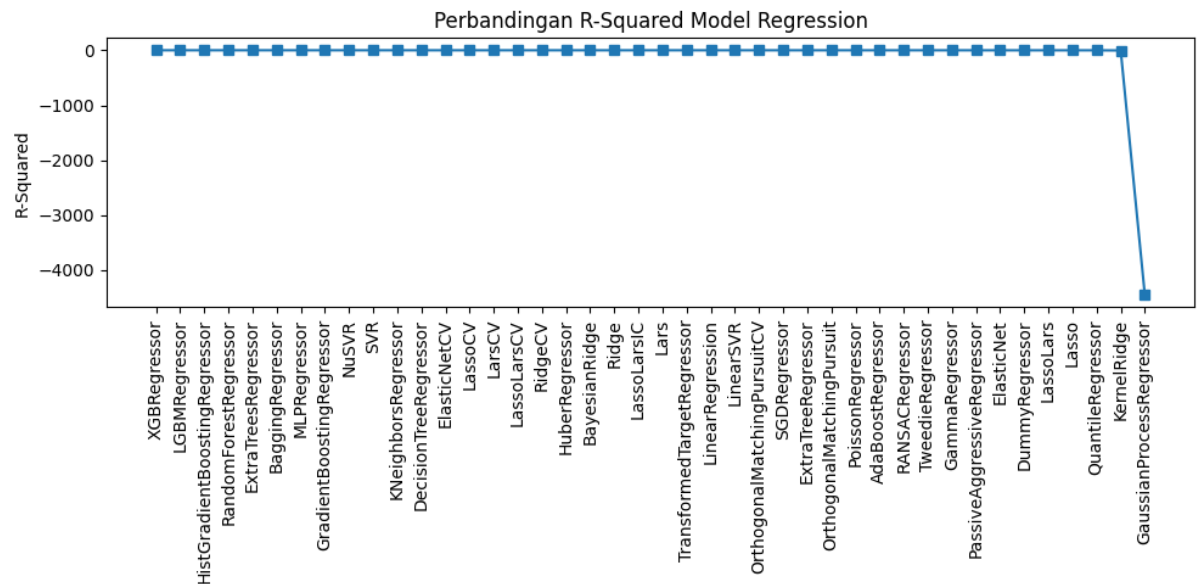
- B

| Model                         | Accuracy | Balanced Accuracy | ROC AUC  | \ |
|-------------------------------|----------|-------------------|----------|---|
| LabelPropagation              | 1.000000 | 1.000000          | 1.000000 |   |
| LinearDiscriminantAnalysis    | 1.000000 | 1.000000          | 1.000000 |   |
| LabelSpreading                | 1.000000 | 1.000000          | 1.000000 |   |
| QuadraticDiscriminantAnalysis | 1.000000 | 1.000000          | 1.000000 |   |
| GaussianNB                    | 0.973684 | 0.979167          | 1.000000 |   |
| ExtraTreesClassifier          | 0.973684 | 0.979167          | 1.000000 |   |
| ExtraTreeClassifier           | 0.973684 | 0.979167          | 0.982759 |   |
| KNeighborsClassifier          | 0.973684 | 0.979167          | 0.998948 |   |
| DecisionTreeClassifier        | 0.973684 | 0.979167          | 0.982759 |   |

- C



- D



➤ OUTPUT BAGIAN EXERCISE

### 1.3.1

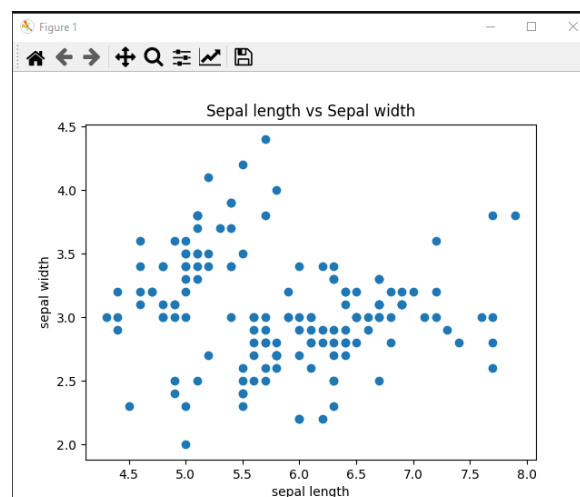
```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\exercise3.1.PY"
[1]

[Done] exited with code=0 in 1.391 seconds
```

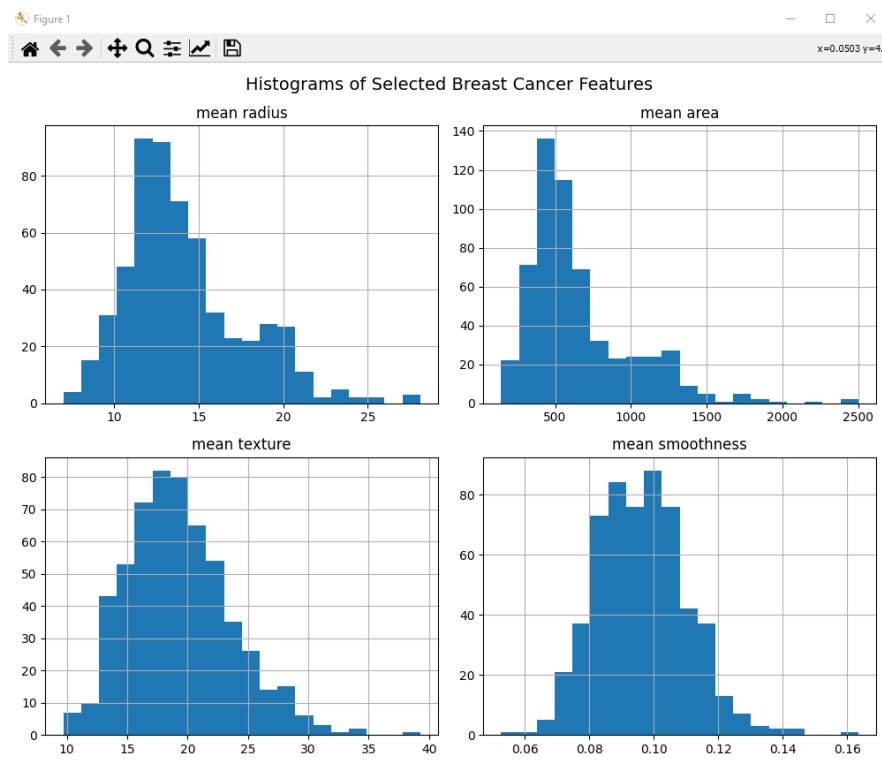
### 2.3.2

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\exercice3.2.PY"  
[0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0  
0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1  
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2  
2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2  
2 2]  
[2]  
  
[Done] exited with code=0 in 1.5 seconds
```

3.3.3.



#### 4.3.4



#### 5.3.5

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\exercise3.5.PY"
[0]

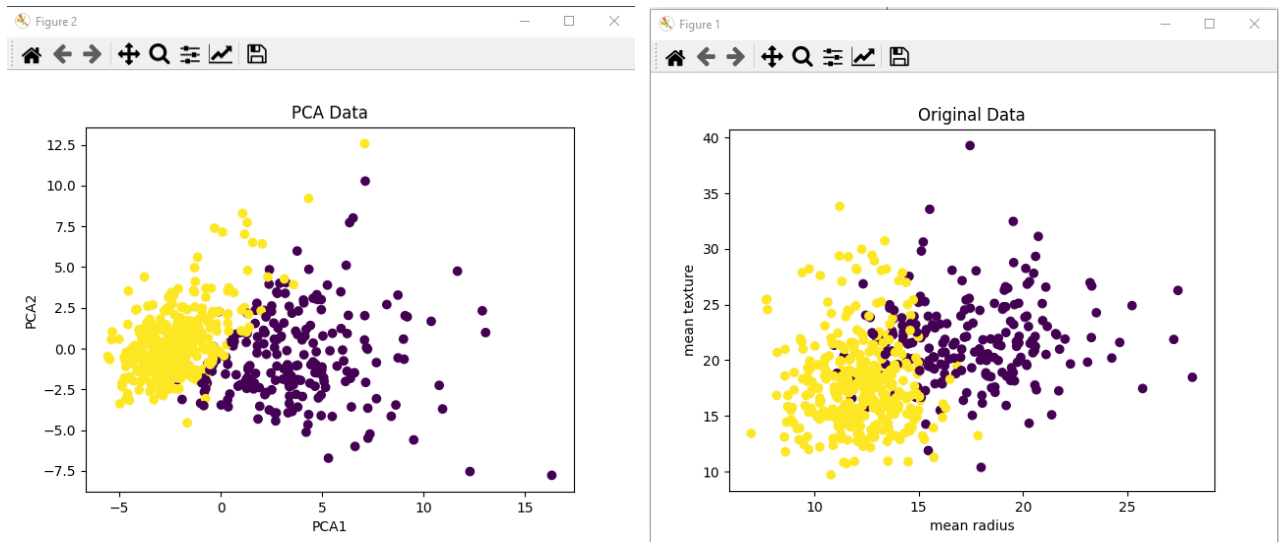
[Done] exited with code=0 in 1.408 seconds
```

#### 6.3.6

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\exercise3.6.PY"
[[-0.07196081 -0.7173124  0.17689245 -1.08801454 -0.25321737  1.09827306]
 [ 0.91090936 -0.51015031 -1.83956679 -0.21142861 -0.22966324  0.18697941]
 [-0.98621136 -0.93180254  0.50377947  1.9103425  0.55378119 -0.58748147]
 ...
 [ 0.65026684  1.22642691  0.95649548  1.58613399  0.79128401 -0.55083369]
 [ 0.30000342  1.06696333 -0.671678  -0.69682734 -1.39615542 -0.56004991]
 [ 0.07241725  1.393738  0.32467708  1.40760881 -0.07983545  0.25152865]]
[0]

[Done] exited with code=0 in 1.534 seconds
```

### 7.3.7



### 8.3.8

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\exercise3.8.PY"
Total points: 89 Correctly labeled points : 80

[Done] exited with code=0 in 1.6 seconds
```

### 9.3.9

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\exercise3.9.PY"
R-squared score on test data: 0.3297

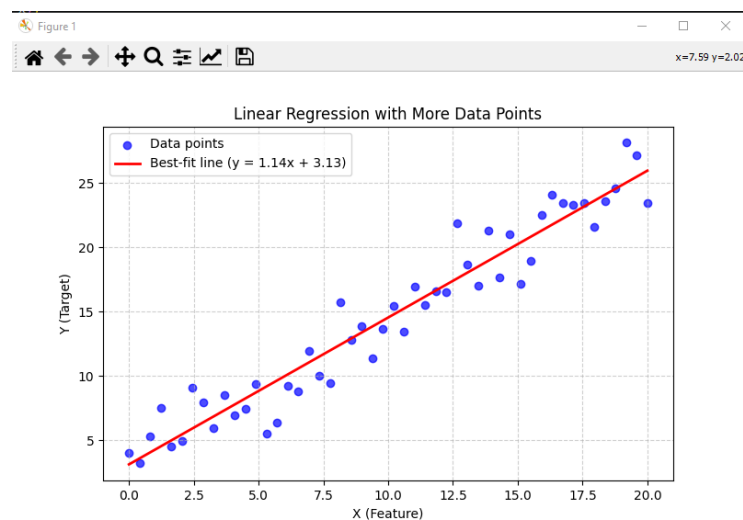
[Done] exited with code=0 in 1.889 seconds
```

### 10.3.10

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\exercise3.10.PY"
SVR: 0.0964032518489577
Linear Regression: 0.43774971182541
Decision Tree: -0.18335302714924162
Random Forest: 0.34297953747805676
Nearest Neighbors: 0.3199078093887773
C:\Users\rifqy\AppData\Local\Programs\Python\Python311\Lib\site-packages\sklearn\normalization\multilayer_perceptron.py:691: ConvergenceWarning:
Stochastic Optimizer: Maximum iterations (2000) reached and the optimization hasn't converged yet.
  warnings.warn(
Neural Networks: 0.4403146277589589

[Done] exited with code=0 in 5.199 seconds
```

### 11.3.11



12. 3.12

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\exercise3.12.PY"
Coefficients:
[[-0.47502636 -0.21771647  0.09308837]
 [-0.13687023 -0.04033662  0.0279736 ]
 [ 0.00107079  0.04202941 -0.02946117]]
Intercept:
[208.23351881  40.59787542  52.04362105]
Predition:
[[205.04753896  39.87056725  52.09059855]]

[Done] exited with code=0 in 1.53 seconds
```

13 3.13

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\exercise3.13.PY"
[0 0 1 1 0 0 1 1 1 1 0 0 1 1 1 0 0 1 1 1 0 1 0 1 1 0 1 0 0 1 1 1 0 1 0 1 0
 0 0 1 0 1 0 1 1 0 1 1 0 0 0 0 0 0 0 0 1 0 1 0 1 1 1 0 1 0 1 0 1 0 0 1 0 1
 0 1 1 0 1 0 0 1 1 0 1 0 0 0 1 0 1 1 1 0 1 0 1 1 1 0 1 0 1 0 0 1 0 1 1 1
 1 1 0 1 1 1 1 1 1 1 1 0 1 1 1 0 0 0 1 0 1 1 0 1 1 0 1 1 1 0 1 0 0 1 0 0 0
 0 1 1 1 0 0 1 1 1 0 0 0 0 0 1 1 0 0 0 1 0 0 0 0 0 1 0 0 1 1 0 1 1 1 1 0 1
 1 0 1 1 0 0 1 0 1 0 0 1 1 0 0 1 0 0 1 1 1 1 1 0 1 1 0 1 1 0 0 0 1 1 1 0
 0 0 0 1 0 0 0 0 0 0 0 0 0 0 1 0 1 0 1 0 0 0 0 1 1 1 1 0 1 0 0 1 1 1 1 0 0
 0 0 1 1 1 0 0 0 0 1 1 1 0 1 1 1 0 0 1 1 0 1 0 0 0 0 1 0 0 0 0 0 1 0 0 1 1
 0 1 0 1]
[[ 1.00426797  4.35026598  1.87395778]
 [ 0.83803951 -1.71264881  2.84821322]]
[1]

[Done] exited with code=0 in 1.677 seconds
```

14. 3.14

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\exercise3.14.PY"
[ 0. -1. -1. -1. -1. -1. -1. -1. -1. -1. -1.  1.]
[0. 0. 0. 0. 0. 0. 1. 1. 1. 1. 1. 1.]

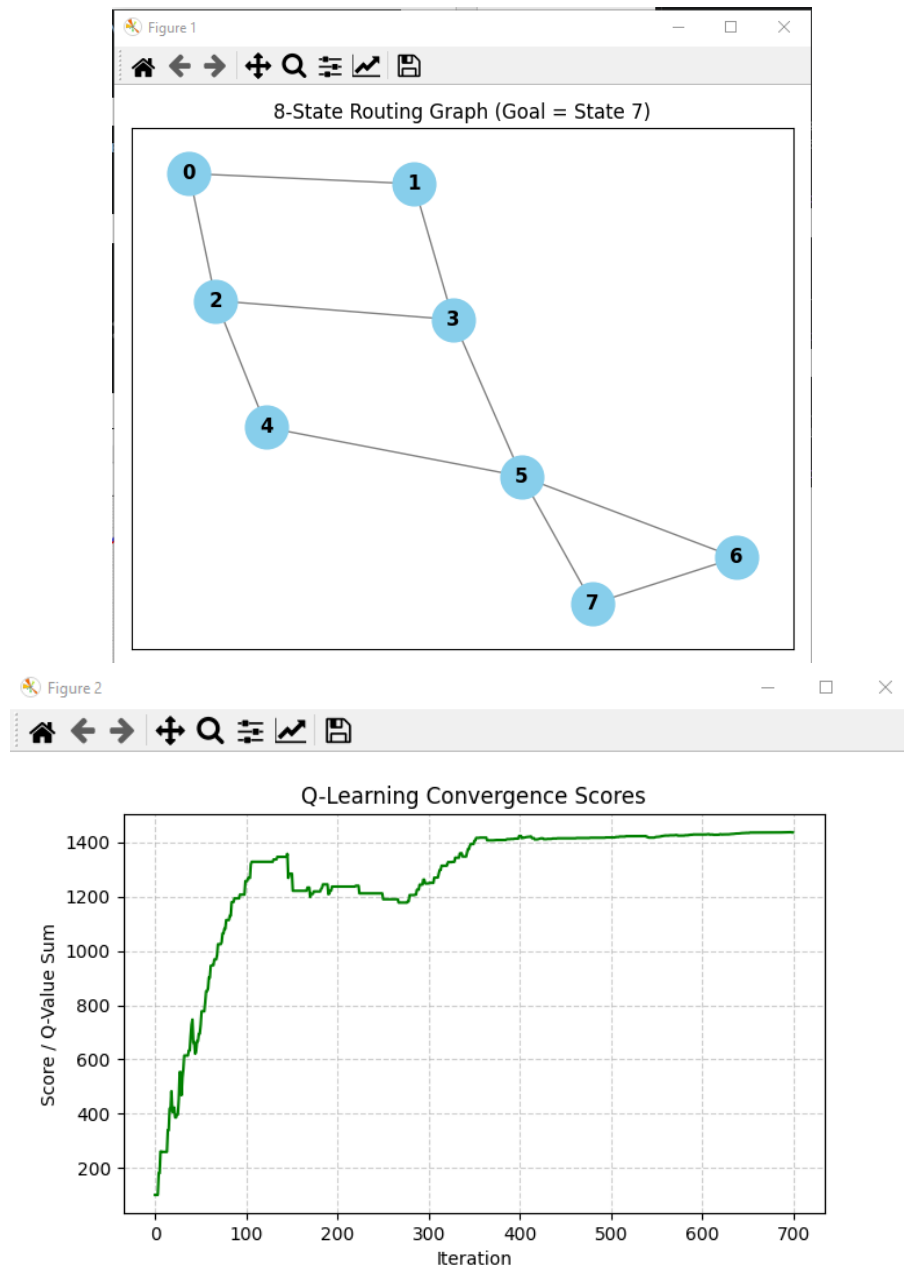
[Done] exited with code=0 in 1.387 seconds
```

15 3.15

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\exercise3.15.PY"
[ 0. -1. -1. -1.  1. -1. -1. -1.  2. -1. -1. -1.]
[0. 0. 0. 0. 1. 1. 1. 1. 2. 2. 2. 2.]

[Done] exited with code=0 in 1.439 seconds
```

16. 3.16



17. 3.17

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\exercise3.17.PY"
Accuracy: 0.95 (+/-0.04) [Logistic Regression]
Accuracy: 0.94 (+/-0.04) [Random Forest]
Accuracy: 0.91 (+/-0.04) [naive Bayes]
Accuracy: 0.95 (+/-0.04) [SVM]
Accuracy: 0.95 (+/-0.04) [KNN]
Accuracy: 0.95 (+/-0.04) [Ensemble]

[Done] exited with code=0 in 2.577 seconds
```

18. 3.18

```
--- Hasil Evaluasi AutoML ---
```

|                               | Adjusted R-Squared | ... | Time Taken |
|-------------------------------|--------------------|-----|------------|
| Model                         |                    | ... |            |
| HistGradientBoostingRegressor | 0.790137           | ... | 0.488456   |
| GradientBoostingRegressor     | 0.768332           | ... | 0.625328   |
| ExtraTreesRegressor           | 0.765534           | ... | 0.697285   |
| MLPRegressor                  | 0.760004           | ... | 1.843559   |
| RandomForestRegressor         | 0.754856           | ... | 1.643919   |
| SVR                           | 0.743385           | ... | 0.223441   |

19. 3.19

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\exercise3.19.PY"
```

|   | Description                 | Value     |
|---|-----------------------------|-----------|
| 0 | Session id                  | 3546      |
| 1 | Target                      | Target    |
| 2 | Target type                 | Binary    |
| 3 | Original data shape         | (569, 31) |
| 4 | Transformed data shape      | (569, 31) |
| 5 | Transformed train set shape | (398, 31) |

20. 3.20

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\exercise3.20.PY"
```

|                               | Accuracy | ... | Time Taken |
|-------------------------------|----------|-----|------------|
| Model                         |          | ... |            |
| LogisticRegression            | 0.979021 | ... | 0.034926   |
| LinearSVC                     | 0.965035 | ... | 0.028980   |
| SVC                           | 0.965035 | ... | 0.029923   |
| PassiveAggressiveClassifier   | 0.965035 | ... | 0.021941   |
| QuadraticDiscriminantAnalysis | 0.965035 | ... | 0.021949   |
| DecisionTreeClassifier        | 0.965035 | ... | 0.027548   |

21. 3.21

```
[Running] python -u "c:\Users\rifqy\OneDrive\TUGAS S5\AI\BAB 3\EXAMPLE\exercise3.21.PY"
```

|                            | Adjusted R-Squared | ... | Time Taken |
|----------------------------|--------------------|-----|------------|
| Model                      |                    | ... |            |
| PassiveAggressiveRegressor | 0.151031           | ... | 0.009974   |
| TransformedTargetRegressor | 0.116748           | ... | 0.010006   |
| LinearRegression           | 0.116748           | ... | 0.009012   |
| ElasticNet                 | 0.115076           | ... | 0.009540   |
| Ridge                      | 0.114627           | ... | 0.009035   |
| ElasticNetCV               | 0.113984           | ... | 0.083850   |
| HuberRegressor             | 0.112070           | ... | 0.027979   |
| BayesianRidge              | 0.112055           | ... | 0.011019   |
| RidgeCV                    | 0.111446           | ... | 0.010009   |
| LassoLarsCV                | 0.104312           | ... | 0.024943   |
| LassoCV                    | 0.104179           | ... | 0.092716   |
| SGDRegressor               | 0.103815           | ... | 0.010006   |